Project Completion Report

Clean Water Well — Nkwen-Mbatu Village

Field	Detail
Reference	ADA/2024/002
Project Title	Borehole Drilling and Solar Pump Installation — Nkwen-Mbatu
Implementing Organization	Apoti Development Association (ADA)
Location	Nkwen-Mbatu Village, ~15 km from Bamenda, North-West Region, Cameroon
Implementation Period	June 1 – October 31, 2024
Total Expenditure	2,650,000 CFA (~$4,417 USD)
Direct Beneficiaries	300+ villagers
Status	✅ Completed

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1. Executive Summary

The Clean Water Well project in Nkwen-Mbatu village was successfully completed within the planned 5-month timeframe and under budget. A 42-meter borehole was drilled, equipped with a solar-powered submersible pump, and connected to an elevated storage tank with two community tap stands. The project provides clean, safe drinking water to over 300 villagers who previously depended on a polluted stream 2 km away.

This was ADA's largest and most technically complex infrastructure project to date. The borehole struck water at 42 meters — 3 meters shallower than the estimated 45 meters — resulting in cost savings of 50,000 CFA on drilling. The solar pump system ensures zero recurring energy costs, and a trained 5-member water committee guarantees community ownership and long-term sustainability.

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2. Background

Nkwen-Mbatu village, home to over 300 people, is located 15 km from Bamenda in Cameroon's North-West Region. Before this project, the community's sole water source was an unprotected stream approximately 2 km from the village center. Women and girls spent 2–4 hours daily collecting water, with girls frequently missing school. Waterborne diseases — typhoid, dysentery, and diarrhea — were common, particularly among children under five.

ADA identified Nkwen-Mbatu through its community outreach network in early 2024. After initial assessments confirmed both the need and the geological feasibility of a borehole, the project was approved as ADA's flagship infrastructure intervention.

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3. Implementation Timeline

Phase 1: Preparation (June 2024)

Community Mobilization

ADA coordinators held three community meetings in the first two weeks of June. Attendance grew from 45 people at the first meeting to over 120 at the third, as word spread about the project. The village chief and quarter heads endorsed the project and helped identify water committee candidates.

Water Committee Formation

A 5-member water committee was selected through community consensus:

- 1 Chairperson (village elder)
- 1 Treasurer (local teacher)
- 1 Secretary (youth representative)
- 2 Members (both women, representing the primary water collectors)

The inclusion of two women was deliberate — recognizing that women bore the greatest burden of water collection and would be the system's primary users.

Geological Survey

A professional hydrogeologist conducted a Vertical Electrical Sounding (VES) survey on June 18–19, testing three potential sites within the village. The survey identified a promising fractured zone at approximately 40–45 meters depth at a location near the village center, with estimated yield of 1.0–1.5 m^3/hour. This site was selected for drilling.

Phase 2: Drilling (July – August 2024)

Site Preparation

The drilling site was cleared and access improved for the drilling rig during the first week of July. The community contributed voluntary labor for site clearing — a strong early sign of ownership.

Borehole Drilling

Drilling commenced on July 15 using a truck-mounted rotary percussion rig. Progress through the geological layers:

- **0–6 m:** Lateritic clay (red-brown, soft)
- **6–18 m:** Weathered basalt (moderate hardness)
- **18–35 m:** Fresh basalt (hard, slow progress)
- **35–42 m:** Fractured granite-gneiss (water-bearing zone)

Water was struck at 42 meters on July 28 — three meters shallower than the estimated 45 m target. The drilling team reported a sudden increase in air-lift yield, confirming entry into a productive fractured zone. Drilling was terminated at 42 m as the yield was sufficient.

Well Development and Testing

- Air-lift development: 6 hours until water ran clear
- Pump test (8 hours): Sustained yield of 1.2 m^3/hour with minimal drawdown
- Recovery test: 95% recovery within 2 hours (indicating excellent aquifer recharge)
- Water quality: Tested clear for E. coli and coliform bacteria; all chemical parameters within WHO guidelines

Casing and Completion

- 6-inch PVC casing installed to full depth
- Stainless steel screen at 35–42 m (aquifer zone)
- Gravel pack around screen section
- Cement grout sanitary seal from surface to 6 m
- Lockable steel wellhead cap installed

Phase 3: Solar Pump Installation (September 2024)

Solar System

- 4 × 100W polycrystalline solar panels mounted on galvanized steel frame (angled at 6° for Bamenda's latitude)
- MPPT controller with dry-run protection installed in weatherproof enclosure
- Submersible solar pump rated for 50 m head, installed at 38 m depth

Storage and Distribution

- 3,000-liter polyethylene elevated tank installed on 4-meter concrete/steel tower
- HDPE rising main from pump to tank
- PVC distribution piping to 2 tap stands
- Concrete aprons with drainage channels at each tap stand
- Brass taps with lockable handles

Testing and Commissioning

The system was tested over a 2-week period in late September. Daily output averaged 4,000–5,000 liters during sunny conditions — more than sufficient for the community's estimated daily demand of 3,000–4,500 liters (based on 10–15 liters per person per day).

Phase 4: Training and Handover (October 2024)

Water Committee Training (October 7–8)

A 2-day intensive training program covered:

- Day 1: System operation, routine maintenance procedures, troubleshooting common issues, hygiene promotion techniques
- Day 2: Financial management, record-keeping, user fee collection, reporting to ADA, conflict resolution

Each committee member received a printed maintenance manual with illustrated troubleshooting guides.

Community Hygiene Sessions

Three community sessions reached approximately 200 adults and school-age children, covering:

- Safe water handling and storage
- Handwashing practices
- Recognizing waterborne disease symptoms
- Proper use of tap stands and drainage

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4. The Day Water Flowed

On October 19, 2024, Nkwen-Mbatu village gathered for the official commissioning ceremony. Over 250 people attended — nearly the entire village — along with the village chief, quarter heads, local government representatives, and ADA's leadership.

The village chief offered a prayer of thanksgiving before the water committee chairperson was invited to open the tap for the first time. As clear water flowed from the tap stand, the crowd erupted in singing and ululation. Women who had spent years walking to the distant stream wept openly. Children ran to touch the water, laughing and splashing.

An elderly woman, one of the village's widows supported by ADA's programs, told the gathering: *"I have carried water on my head since I was a girl. My daughters carried water. My granddaughters carried water. Today, the water has come to us."*

The village chief declared the day a community holiday. Families brought food, and the celebration continued into the evening with traditional music and dancing. Several mothers spoke about what the time savings would mean — their daughters could now attend school regularly, and they themselves could pursue small trading activities in the mornings previously lost to water collection.

The ADA team documented the celebration extensively. The joy was palpable — this was not merely infrastructure, but liberation from a burden carried for generations.

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5. Outcomes Achieved

Indicator	Target	Achieved
People with clean water access	300+	300+ (entire village)
Borehole drilled and functional	1	1 (42 m depth, 1.2 m^3/hr yield)
Solar pump system installed	1	1 (400W, zero recurring cost)
Water committee trained	5 members	5 members (2-day program)
Tap stands operational	2	2
Daily water output	3,000+ liters	4,000–5,000 liters

Secondary Outcomes

- **Time savings:** Women and girls save 2–4 hours daily previously spent on water collection
- **School attendance:** Early reports indicate improved attendance among girls
- **Health impact:** No reported cases of waterborne disease in the first month post-commissioning
- **Economic opportunity:** Women report using saved time for farming and small trade
- **Community cohesion:** The project strengthened community organization and collective action capacity

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6. Challenges and Lessons Learned

Challenges Encountered

1. **Access road condition:** Heavy rains in July made the access road difficult for the drilling rig. Resolved by community labor to improve the road surface.
2. **Drilling rig scheduling:** Initial contractor availability delayed start by 2 weeks. Mitigated by parallel community mobilization activities.
3. **Solar panel transport:** Fragile panels required careful transport on rough roads. No damage occurred but highlighted need for protective packaging.

Lessons Learned

1. **Community engagement is foundational.** Voluntary labor for site clearing demonstrated that early, genuine engagement builds ownership.
2. **Geological surveys save money.** The 150,000 CFA investment ensured drilling at the optimal location, hitting water 3 m shallower than estimated.
3. **Women's inclusion matters.** Having women on the water committee ensures the system serves its primary users.
4. **Solar is ideal for rural Cameroon.** Zero recurring costs, minimal maintenance, and sufficient sunshine year-round.
5. **Infrastructure projects build organizational capacity.** This project stretched ADA's capabilities and established processes for future work.

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7. Recommendations

1. **6-month follow-up visit** to assess system performance, water quality, and committee functioning
2. **Annual water quality testing** to ensure continued safety
3. **Replicate model** in neighboring villages with similar geological conditions
4. **Document as case study** for future fundraising and donor engagement
5. **Connect water committee** to regional WASH networks for peer learning

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8. Conclusion

The Clean Water Well project in Nkwen-Mbatu represents a transformative intervention for a community that endured generations without safe water access. Delivered on time and under budget, the project demonstrates that targeted infrastructure investment, combined with genuine community engagement and appropriate technology, can produce lasting change.

The image of clear water flowing from the tap on October 19 will remain a defining moment for ADA and for Nkwen-Mbatu village. Water is life, and life has come to Nkwen-Mbatu.

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